



# University of Asia Pacific

Department of Computer Science and Engineering

## CSE 316: Microprocessors and Microcontrollers Lab

### LAB REPORT

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**Experiment Number:** 08

**Experiment Title:** Automated Plant Watering System

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## 1. Experiment Name

Automated Plant Watering System.

## 2. Objective

This project focuses on designing a system that can automatically monitor soil moisture level and control a water pump using Arduino. The system detects whether the soil is dry or wet and activates the motor accordingly. **LEDs** are used to indicate the soil condition visually.

## 3. Apparatus / Hardware & Software Requirements

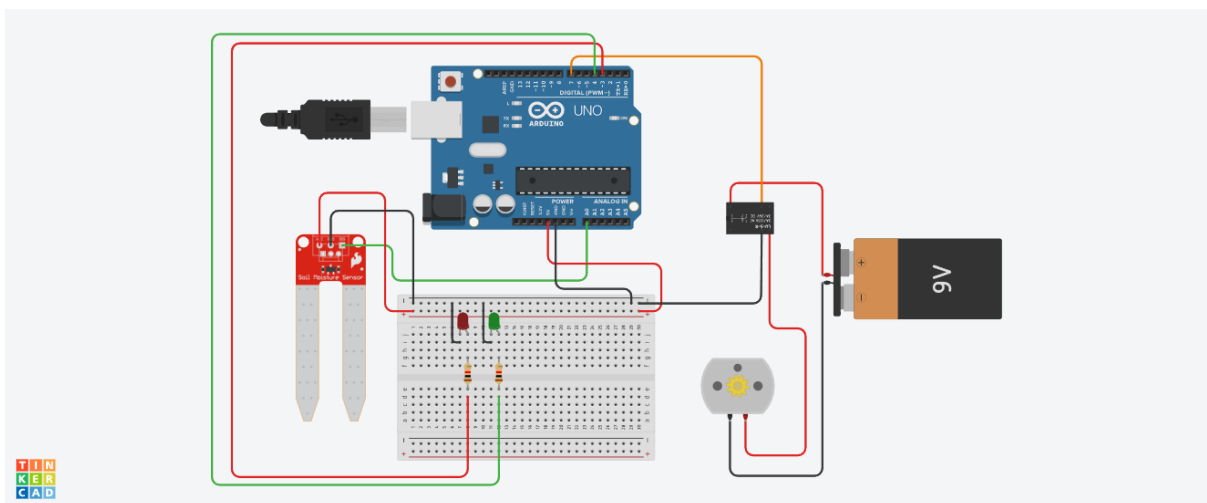
### Hardware Components:

- Arduino Uno R3
- Soil Moisture Sensor
- Relay Module
- DC Motor / Water Pump
- Red LED
- Green LED
- 1k $\Omega$  Resistors
- 9V Battery
- Breadboard
- Jumper Wires

### Software:

- Arduino IDE
- Tinkercad (for simulation)

## 4. Circuit Diagram / Schematic



**Figure - 01:** Automated Plant Watering System.

## 5. Code / Assembly Program

```
const int soilPin = A0;
const int relayPin = 7;

const int redLED = 3;
const int greenLED = 4;

int moistureValue = 0;

// Moisture threshold value
int threshold = 500;

void setup() {

    Serial.begin(9600);

    pinMode(relayPin, OUTPUT);

    pinMode(redLED, OUTPUT);
    pinMode(greenLED, OUTPUT);

    Serial.println("Soil Moisture Monitoring System Started...");
}

void loop() {

    moistureValue = analogRead(soilPin);

    Serial.print("Moisture Value: ");
    Serial.println(moistureValue);
```

```
// Soil is Dry
if (moistureValue > threshold) {

    Serial.println("Soil is Dry - Motor ON");

    digitalWrite(relayPin, HIGH);

    digitalWrite(redLED, HIGH);
    digitalWrite(greenLED, LOW);
}

// Soil is Wet
else {

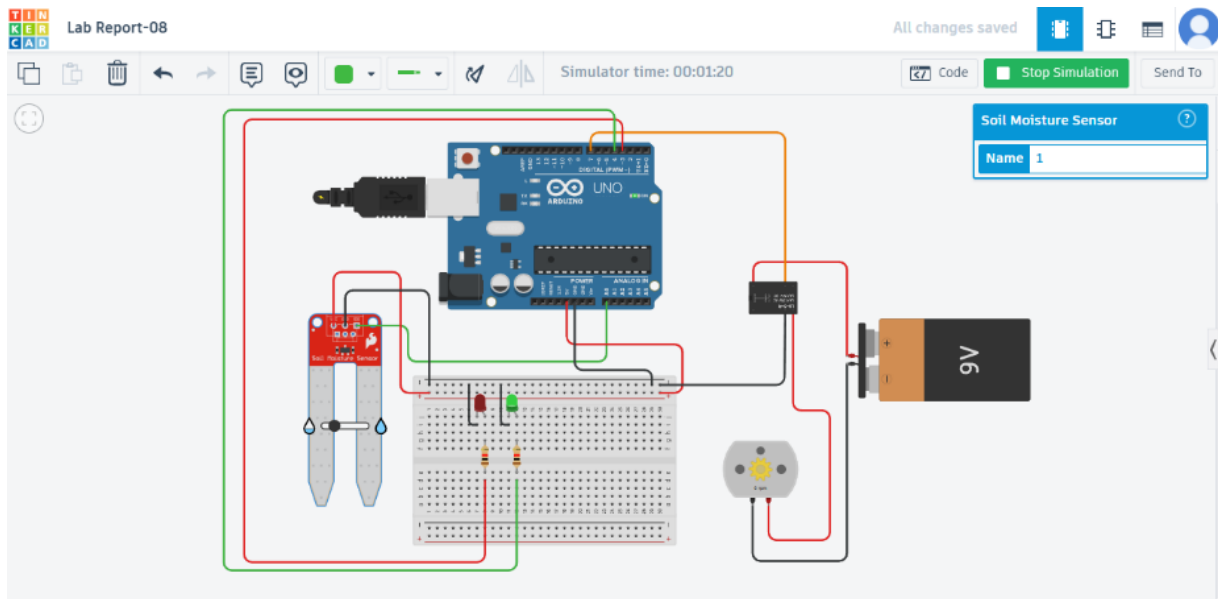
    Serial.println("Soil is Wet - Motor OFF");

    digitalWrite(relayPin, LOW);

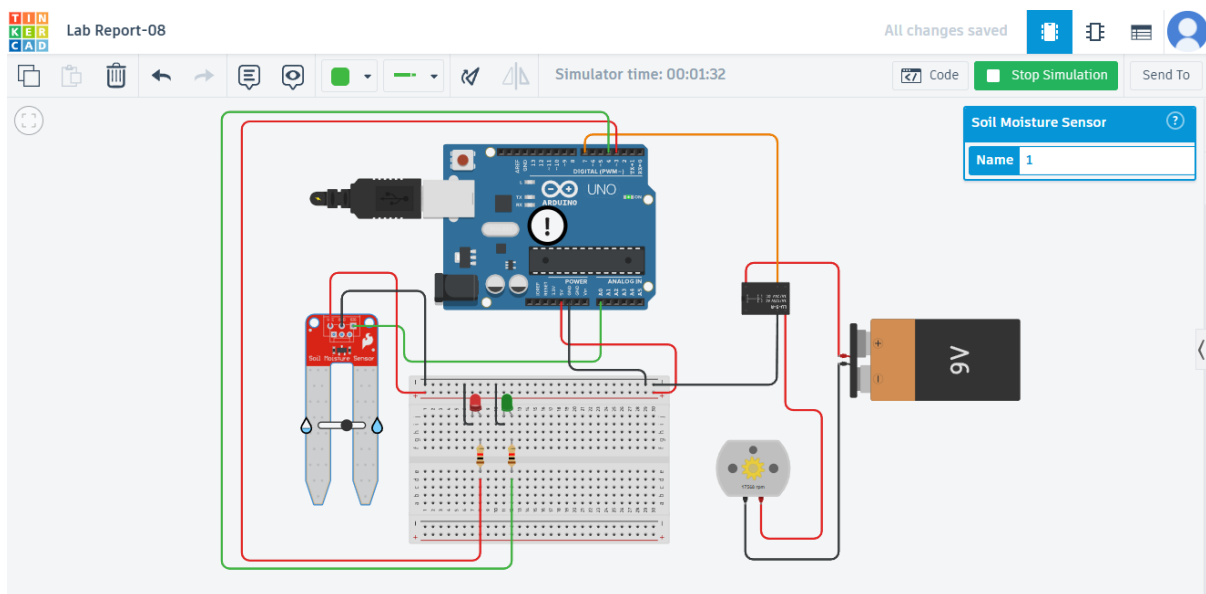
    digitalWrite(redLED, LOW);
    digitalWrite(greenLED, HIGH);
}

delay(1000);
}
```

## 6. Output / Observations



**Figure-02:** Soil Wet, Green LED = ON.



**Figure-03:** Soil Dry, Red LED = ON.

- The soil moisture sensor continuously monitors the moisture level of the soil.
- When the soil becomes dry, the sensor sends a signal to the Arduino.
- Arduino activates the relay which turns **ON** the **DC motor** (water pump).
- Triggering the relay allows current from the battery to power the motor.
- During **dry** condition, the **Red LED** glows.
- When sufficient moisture is detected, the relay turns **OFF** and stops the motor.
- During **wet** condition, the **Green LED** glows.
- The system gives automatic response according to the moisture level.

## 7. Result

The soil moisture monitoring system was successfully designed and implemented. The system managed to detect soil moisture conditions and controlled the water pump automatically through the relay module. The **LEDs** successfully indicated dry and wet soil conditions. The objectives of the experiment were accomplished successfully.

## 8. Conclusion

By combining an Arduino microcontroller with real-time moisture sensors, this experiment bridges the gap between traditional farming and modern tech. It illustrates how simple components like relays, indicators, and code can unite to form an autonomous irrigation network—a foundational step toward resource-efficient, smart agricultural ecosystems.